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[13AE52]

# FIFTH SEMESTER B.TECH EXAMINATION, DECEMBER– 2015

## AERODYNAMICS - II

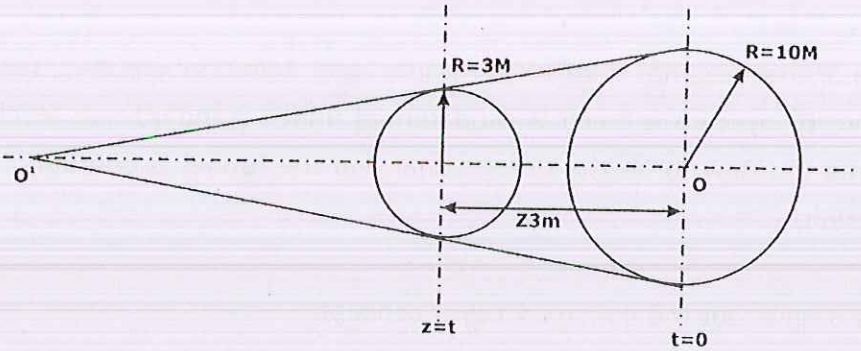
**Time: 3 Hours** **Maximum Marks:70**

**Instructions:**

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

**1. a.** Define compressibility and prove that 'isentropic' compressibility is smaller than isothermal compressibility by the factor $\gamma$ (ratio of specific heads). **06**

**b.** Two consecutive sound disturbances spheres created by a supersonic plane are shown in the Fig. 1. Assuming medium to be standard sea level atmosphere calculate the Mach no. velocity of plane and Mach angle. (* in M/sec) **08**




**2. a.** Derive one dimensional momentum equation (Euler's) for inviscid compressible flow. **06**

**b.** Consider isentropic flow through a rocket engine nozzle. In the combustion chamber, the gas is at a temperature and preserve of 15atm and 2500K. The molecular weight and specific heat at context pressure of combustion gas are 12 and 4157J/Kg K. The gas expands to supersonic speed through the nozzle with a temperature of 1350K at the nozzle exit. Calculate the pressure, the Mach number and velocity at the exit of the nozzle. (pressure in atm., velocity in m/s) **08**

**3. a.** Derive velocity of a moving shock have as a function of pressure ratio across the shock and the speed of sound. **06**

**b.** A normal shock is propagating into stagnant air where the ambient temperature is 310K and the pressure ratio is 10. Calculate the moving shock wave velocity and the induced mass motion velocity. **08**

**4.** For the flow over a diamond shaped air foil with 30° included angle and zero degree angle of attack with M=1.8, find the pressure co-efficient over the top surfaces. **14**

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| **5.** | **a.** | Derive area-Mach number relation for an isentropic flow through a convergent-divergent nozzle. | **06** |
| | **b.** | For different pressure ratios explain with neat sketches the flow through a convergent divergent nozzle. (Including over-expansion, correct expansion and under expansion flows). | **08** |
| **6.** | **a.** | From the expanded form of the perturbation velocity potential equation deduce linearized small perturbation equation for subsonic and supersonic flows. | **06** |
| | **b.** | Deduce linearized lift and drag coefficients ($C_L$ and $C_D$) for a flat plate at small angle of attack $(\alpha)$ in a supersonic flow. | **08** |
| **7.** | **a.** | Draw a neat sketch of a blow-down supersonic wind tunnel and mention the functions of each unit. State the advantages and disadvantages over other tunnels. | **06** |
| | **b.** | Estimate the settling chamber pressure and temperature and the area ratio required to operate a Mach 2 wind tunnel under standard sea level conditions. (Assume the flow to be the dimensional and the tunnel is operating with correct expansion. | **08** |
| **8.** | **a.** | Give a brief about the boundary Layer concept. | **06** |
| | **b.** | What are the important consequences, fundamental for boundary Layer theory? | **08** |